Note of Fast Accurate Eigenvalue Computations Using The Cholesky Factorization

Theorem 2.3:
Let $G = BD \in \mathbb{R}^{n\times n}$ be nonsingular and let $\hat{G} = (B + \Delta B)D$ where $\|\Delta B\| \le \sigma_n(B)$.

$$\left|\frac{\sigma_i(\hat{G}) - \sigma_i(G)}{\sigma_i(G)}\right| \le \frac{\|\Delta B\|}{\sigma_n(B)}, \quad i = 1, \ldots, n.$$

Co

$$G = L\tilde{D}^{-1}\tilde{D}$$

$$\tilde{G} = (L\tilde{D}^{-1} - \Delta\tilde{D}^{-1})\tilde{D} = L_{(1)}\tilde{D}^{-1}\tilde{D}$$

$$\left\|L\tilde{D}^{-1}\right\| \le \frac{\left\|\Delta\tilde{D}^{-1}\right\|}{\sigma_n(L\tilde{D}^{-1})}$$

1 Derivative of eigenvalues and eigenvectors

Derivative of an eigenvalue $\lambda(X)$ and an eigenvector $u(X)$ of matrix X at X_0.
$\cdot^+$ means Moore-Penrose inverse.

Real symmatric case $X_0 \in \mathbb{R}^{n\times n}$, $X_0 = X_0^T$:

$$X_0 u_0 = \lambda_0 u_0$$

and λ_0 be a simple eigenvalue, then

$$Xu = \lambda u, \quad u^T u = 1, \quad X \in N(X_0)$$

$$\mathrm{d}\lambda = u_0^T(\mathrm{d}X)u_0$$
$$\mathrm{d}u = (\lambda_0 I_n - X_0)^+(\mathrm{d}X)u_0$$

General case $Z_0 \in \mathbb{C}^{n\times n}$:

$$Z_0 u_0 = \lambda_0 u_0$$

and λ_0 be simple eigenvalue, then

$$Zu = \lambda u, \quad u_0^* u = 1, \quad Z \in N(Z_0)$$

$$\mathrm{d}\lambda = \frac{v_0^*(\mathrm{d}Z)u_0}{v_0^* u_0}$$

$$\mathrm{d}u = (\lambda_0 I - Z_0)^+ \left(I - \frac{u_0 v_0^*}{v_0^* u_0}\right)(\mathrm{d}Z)u_0$$

where $Z_0^* v_0 = \bar{\lambda}_0 v_0$, i.e. v_0^* is the left eigenvector of Z_0 corresponding to λ_0.